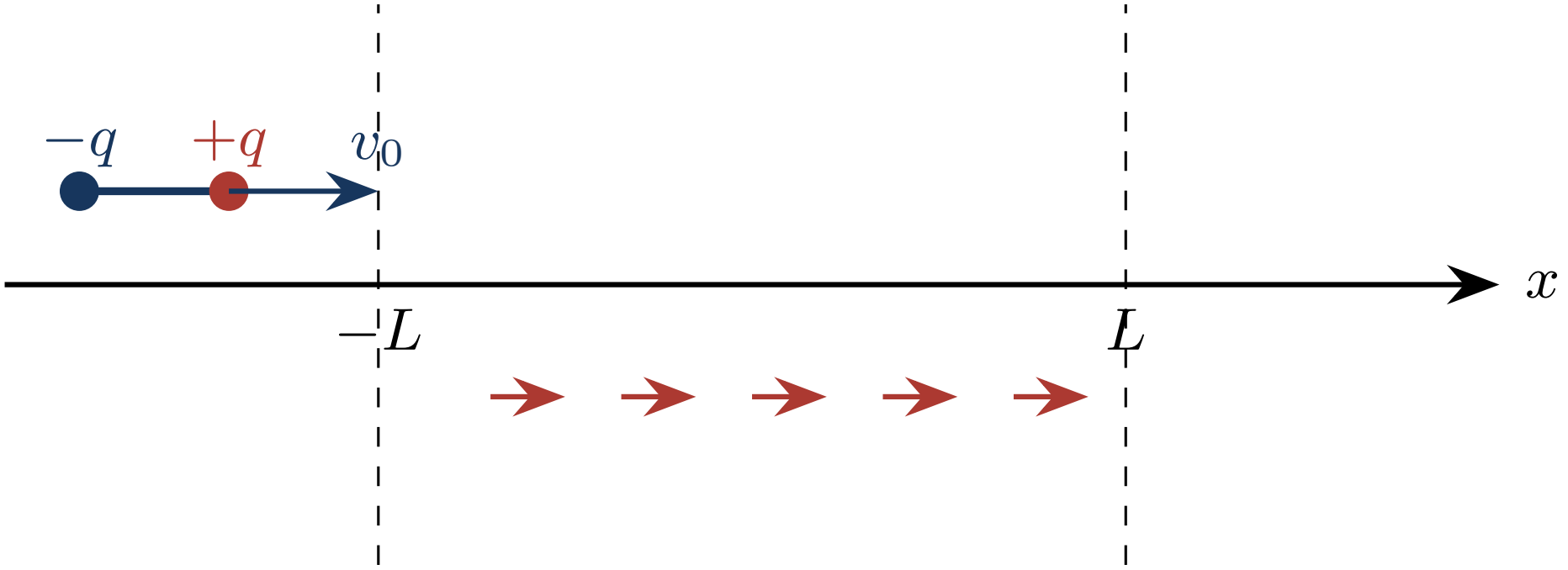
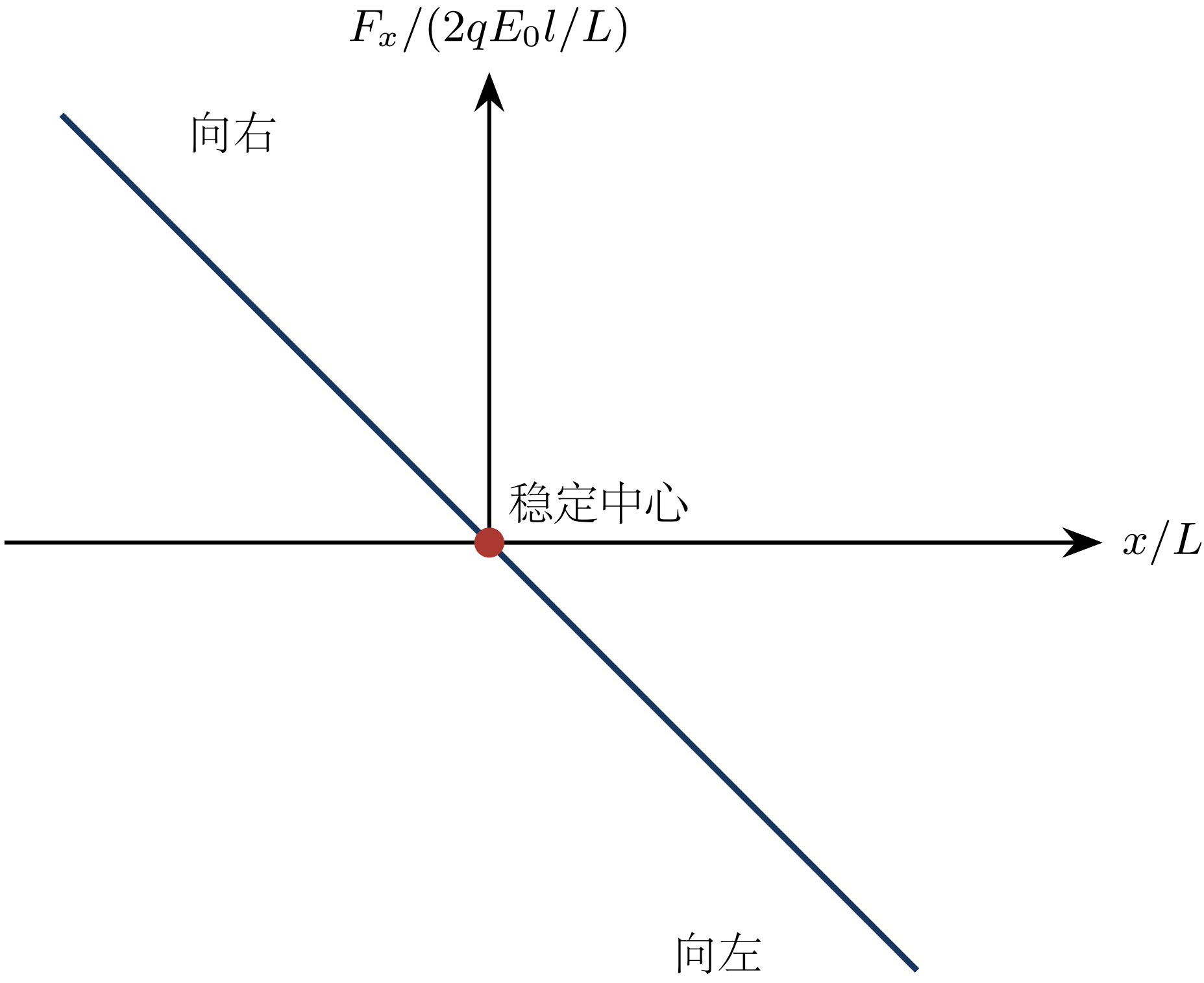
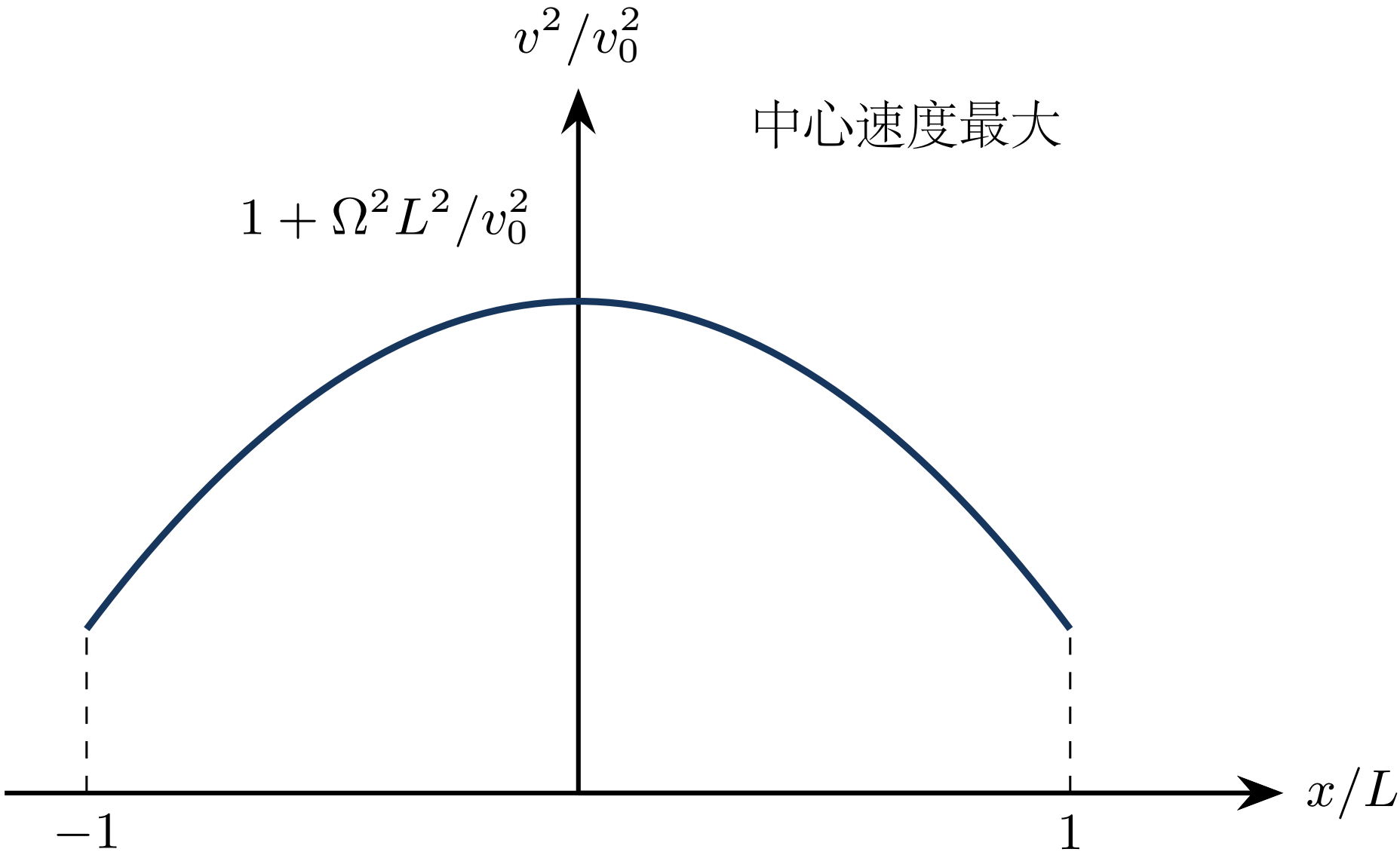
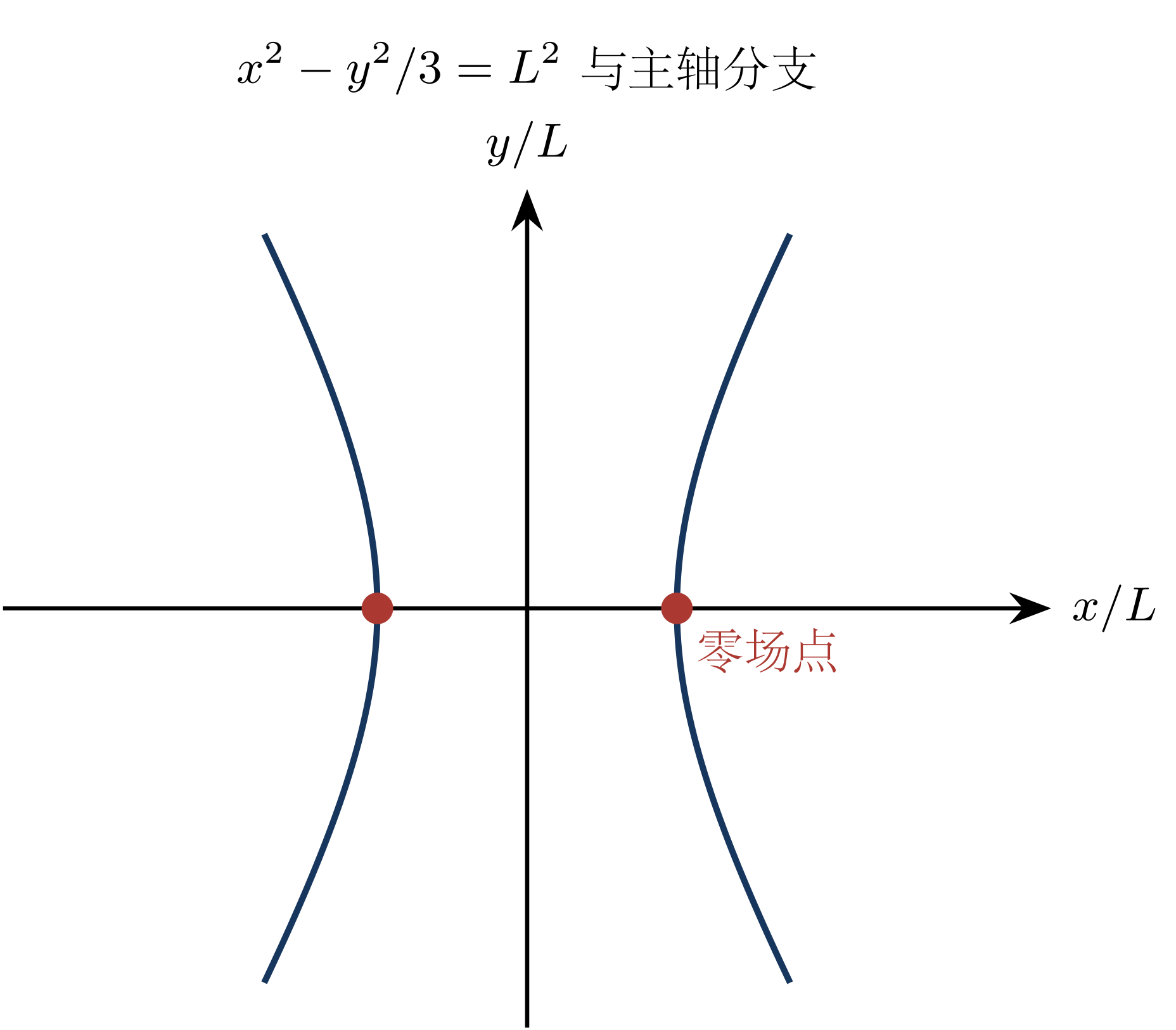
主轴 $E_x = E_0(1 - x^2/L^2)$




$F_x/(2qE_0l/L)$
向右
稳定中心
x/L
向左


v^2/v_0^2
中心速度最大
$1+\Omega^2 L^2/v_0^2$
x/L
-1
1


$x^2 - y^2/3 = L^2$ 与主轴分支
y/L
x/L
零场点

主轴场 $E_x(x,0)$

无电荷静电延拓
关于 x 轴镜面对称

$$E_x = E_0(1 - x^2/L^2 + y^2/L^2)$$
$$E_y = 2E_0 xy/L^2$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0, \quad \boldsymbol{\nabla} \times \boldsymbol{E} = 0$$